

DBMS – SELLER AND INVENTORY MANAGEMENT

BOOKSWAGON – E-Commerce Order Management Database System

1. DATABASE DESIGN

1.1 SELLER TABLE

Attribute	Description
Seller_ID	Unique seller ID
Seller_Name	Seller name
Contact_Number	Seller contact number
Email	Seller email address
Address	Seller address

1.2 INVENTORY TABLE

Attribute	Description
Inventory_ID	Unique inventory ID
Product_ID	Product ID
Seller_ID	Seller ID
Stock_Quantity	Available stock quantity
Stock_Status	Availability status of the product
Last_Updated	Date and time when inventory was last updated

2. TABLE RELATIONSHIP

3. TABLE CREATION

3.1 SELLER TABLE

```
SQL> CREATE TABLE BookswagonSeller (  
  2     Seller_ID NUMBER PRIMARY KEY,  
  3     Seller_Name VARCHAR2(100),  
  4     Contact_Number VARCHAR2(15),  
  5     Email VARCHAR2(100)  
  6 );
```

Table created.

SQL>

```
SQL> ALTER TABLE BookswagonSeller  
  2 ADD Address VARCHAR2(200);
```

Table altered.

3.2 INVENTORY TABLE

```
SQL> CREATE TABLE BookswagonInventory (  
  2     Inventory_ID NUMBER PRIMARY KEY,  
  3     Product_ID NUMBER REFERENCES BookswagonProduct(Product_ID),  
  4     Seller_ID NUMBER REFERENCES BookswagonSeller(Seller_ID),  
  5     Stock_Quantity NUMBER NOT NULL,  
  6     Stock_Status VARCHAR2(20) NOT NULL,  
  7     Last_Updated DATE NOT NULL  
  8 );
```

Table created.

4. SELLER DATA INSERTION

```
SQL> INSERT INTO BookswagonSeller VALUES  
  2 (1, 'Book World', '9876543210', 'bookworld@gmail.com', 'Chennai');
```

1 row created.

SQL>

```
SQL> INSERT INTO BookswagonSeller VALUES  
  2 (2, 'Readers Hub', '9876543211', 'readershut@gmail.com', 'Tambaram');
```

1 row created.

SQL>

```
SQL> INSERT INTO BookswagonSeller VALUES  
  2 (3, 'Novel House', '9876543212', 'novelhouse@gmail.com', 'Velachery');
```

1 row created.

SQL>

```
SQL> INSERT INTO BookswagonSeller VALUES  
  2 (4, 'Book Paradise', '9876543213', 'bookparadise@gmail.com', 'Adyar');
```

1 row created.

SQL>

```
SQL> INSERT INTO BookswagonSeller VALUES  
  2 (5, 'Readers Point', '9876543214', 'readerspoint@gmail.com', 'Chromepet');
```

1 row created.

5. INVENTORY DATA INSERTION

```
SQL> INSERT INTO BookswagonInventory VALUES
      2 (1, 101, 1, 50, 'Available', TO_DATE('18-09-2026','DD-MM-YYYY'));
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO BookswagonInventory VALUES
      2 (2, 102, 2, 80, 'Available', TO_DATE('18-09-2026','DD-MM-YYYY'));
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO BookswagonInventory VALUES
      2 (3, 103, 3, 25, 'Available', TO_DATE('18-09-2026','DD-MM-YYYY'));
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO BookswagonInventory VALUES
      2 (4, 104, 4, 40, 'Available', TO_DATE('18-09-2026','DD-MM-YYYY'));
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO BookswagonInventory VALUES
      2 (5, 105, 5, 60, 'Available', TO_DATE('18-09-2026','DD-MM-YYYY'));
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO BookswagonInventory VALUES
      2 (8, 108, 8, 20, 'Available', TO_DATE('18-09-2026','DD-MM-YYYY'));
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO BookswagonInventory VALUES
      2 (9, 109, 1, 55, 'Available', TO_DATE('18-09-2026','DD-MM-YYYY'));
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO BookswagonInventory VALUES
      2 (10, 110, 2, 0, 'Unavailable', TO_DATE('18-09-2026','DD-MM-YYYY'));
```

1 row created.

6. SELLER PRODUCT INFORMATION

6.1 Seller and Product Details

```
SQL> SELECT
 2     s.Seller_ID,
 3     s.Seller_Name,
 4     p.Product_ID,
 5     p.Product_Name,
 6     p.Price,
 7     i.Stock_Quantity,
 8     i.Stock_Status
 9 FROM BookswagonSeller s
10 JOIN BookswagonInventory i
11 ON s.Seller_ID = i.Seller_ID
12 JOIN BookswagonProduct p
13 ON i.Product_ID = p.Product_ID
14 ORDER BY s.Seller_ID;
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
1
Book World
101
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
The Alchemist
450          50 Available

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
1
Book World
109
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
The Power of Your Subconscious Mind
275 55 Available
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
2
Readers Hub
102
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
Atomic Habits
499 80 Available
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
2
Readers Hub
110
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
Pride and Prejudice
    325          0 Unavailable
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
    3
Novel House
    103
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
The Psychology of Money
    350          25 Available
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE STOCK_QUANTITY STOCK_STATUS
-----
    4
Book Paradise
    104
```

```

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
      PRICE STOCK_QUANTITY STOCK_STATUS
-----
Wings of Fire
      299           40 Available

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
      PRICE STOCK_QUANTITY STOCK_STATUS
-----
              5
Readers Point
          105

```

```

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
      PRICE STOCK_QUANTITY STOCK_STATUS
-----
Harry Potter and the Philosopher Stone
      450           60 Available

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
      PRICE STOCK_QUANTITY STOCK_STATUS
-----
              6
Classic Books Store
          106

```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
      PRICE STOCK_QUANTITY STOCK_STATUS
-----
Rich Dad Poor Dad
      399           30 Available
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
      PRICE STOCK_QUANTITY STOCK_STATUS
-----
              8
Book Basket
      108
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
      PRICE STOCK_QUANTITY STOCK_STATUS
-----
Think and Grow Rich
      250           20 Available
```

9 rows selected.

7. INVENTORY STATUS

7.1 Display Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM BookswagonInventory i
  8 JOIN BookswagonProduct p
  9 ON i.Product_ID = p.Product_ID
 10 JOIN BookswagonSeller s
 11 ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Status = 'Available';
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
      101
The Alchemist
Book World
      50 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
      102
Atomic Habits
Readers Hub
      80 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
      103
The Psychology of Money
Novel House
      25 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
          104
Wings of Fire
Book Paradise
          40 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
          105
Harry Potter and the Philosopher Stone
Readers Point
          60 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
          106
Rich Dad Poor Dad
Classic Books Store
          30 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
          108
Think and Grow Rich
Book Basket
          20 Available
```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
      109
The Power of Your Subconscious Mind
Book World
      55 Available

8 rows selected.

```

7.2 Display Unavailable Products

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM BookswagonInventory i
  8 JOIN BookswagonProduct p
  9 ON i.Product_ID = p.Product_ID
 10 JOIN BookswagonSeller s
 11 ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Status = 'Unavailable';

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
      110
Pride and Prejudice
Readers Hub
      0 Unavailable

```

8. Available and Unavailable Product Count

```

SQL> SELECT
  2     Stock_Status,
  3     COUNT(*) AS Total_Products
  4 FROM BookswagonInventory
  5 GROUP BY Stock_Status
  6 ORDER BY Stock_Status;

STOCK_STATUS      TOTAL_PRODUCTS
-----
Available          8
Unavailable        1

```

9. Total Stock by Seller

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     SUM(i.Stock_Quantity) AS Total_Stock
  5 FROM BookswagonSeller s
  6 JOIN BookswagonInventory i
  7 ON s.Seller_ID = i.Seller_ID
  8 GROUP BY s.Seller_ID, s.Seller_Name
  9 ORDER BY s.Seller_ID;
```

SELLER_ID	SELLER_NAME	TOTAL_STOCK
1	Book World	105
2	Readers Hub	80

SELLER_ID	SELLER_NAME	TOTAL_STOCK
3	Novel House	25
4	Book Paradise	

SELLER_ID	SELLER_NAME	TOTAL_STOCK
40		
5	Readers Point	60
6		

```

SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
-----
Classic Books Store
      30

      8
Book Basket
      20

7 rows selected.

```

10. INVENTORY SUMMARY

```

SQL> SELECT
  2     COUNT(*) AS Total_Inventory,
  3     SUM(CASE
  4         WHEN Stock_Status = 'Available' THEN 1
  5         ELSE 0
  6     END) AS Available_Products,
  7     SUM(CASE
  8         WHEN Stock_Status = 'Unavailable' THEN 1
  9         ELSE 0
 10     END) AS Unavailable_Products,
 11     SUM(Stock_Quantity) AS Total_Stock
 12 FROM BookswagonInventory;

TOTAL_INVENTORY  AVAILABLE_PRODUCTS  UNAVAILABLE_PRODUCTS  TOTAL_STOCK
-----
              9              8              1             360

```